

Factor-Aware Sparse-to-Full Transfer for Intrinsic Riesz Identification

Growing-Horizon Hardy Robustness, a Conditional Closure Theorem, and a Nonnormal No-Go

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Abstract

The small-noise intrinsic-identification program for a folded-Gaussian quadratic transfer operator has reduced the continuum defect to two directional Hardy energies. Previous work supplied deterministic finite-matrix tail certificates and an adaptive sparse approximation, but did not propagate the cutoff through the matrix's own Riesz factors and normalized Haar coupling ranges. We close that finite-dimensional interface.

First, a Hilbert–Schmidt coupling perturbation $\|B - \tilde{B}\|_{\mathfrak{S}_2} \leq \varepsilon_B < \|B\|_{\mathfrak{S}_2}$ gives

$$\left\| \frac{B}{\|B\|_{\mathfrak{S}_2}} - \frac{\tilde{B}}{\|\tilde{B}\|_{\mathfrak{S}_2}} \right\|_{\mathfrak{S}_2} \leq \frac{2\varepsilon_B}{\|B\|_{\mathfrak{S}_2}}.$$

Since adjacent folded-Gaussian couplings have size $\Theta(h\sigma^{-3/2})$, an adaptive cutoff error of order

$$\frac{h^2}{(\log(1/h))^{1/4}}$$

is automatically negligible after normalization. Second, we give explicit left/right defect formulas for the intrinsic triples in terms of the Markov, unweighted-projector, weighted-Riesz, and coupling defects. Third, actual finite-time ledgers $a_j = \|A^j\|$ and $\tilde{a}_j = \|\tilde{A}^j\|$ yield

$$d_M = \delta_A \sum_{j=0}^{M-1} a_{M-1-j} \tilde{a}_j, \quad q_M + d_M < 1,$$

which transports a growing-horizon block contraction and a complete Hardy upper without replacing nonnormal transients by powers of $\|A\|$.

The resulting synthesis is exact but conditional: if the dyadic left and right Hardy budgets are $O(\sigma^{-\alpha_B})$ and $O(\sigma^{-\alpha_C})$, with range-restricted residues bounded, then

$$\|\mathcal{I}_{n,\sigma}\|_{\mathfrak{S}_2} = O\left(n^{-2}\sigma^{-13/4-\alpha_B-\alpha_C}\right).$$

Every strict $n\sigma^2 \rightarrow \infty$ schedule survives when $\alpha_B + \alpha_C \leq 1/4$; polylogarithmic budgets give the expected $n^{-2}\sigma^{-13/4}$ polylog($1/\sigma$) law. We also prove that small matrix error alone cannot control nonnormal Riesz factors, so a contour-conditioning or strong-weak stability ledger is indispensable. A five-scale dense audit through dimension 512 and a 256-bit Arb arithmetic audit confirm the transfer mechanism, but do not prove the remaining dyadically uniform Hardy or Riesz-conditioning premises.

Keywords: Riesz projection; nonnormal perturbation; Hilbert–Schmidt Hardy energy; growing horizon; Gaussian cutoff; intrinsic Galerkin identification.

MSC 2020: 47A10; 47B10; 65F35; 65G20; 93B36.

1 Introduction

For each positive noise width σ , the folded-Gaussian quadratic operator has two simple peripheral branches: the Perron root and a negative parity resonance approaching -1 . Subtracting their intrinsic weighted Riesz terms produces the bulk relevant to the two-step trace and determinant program. The continuum terms themselves are spatially resolved under the strict mesh condition $n\sigma^2 \rightarrow \infty$, but a finite Galerkin matrix recomputes its own spectral terms. The remaining defect is therefore

$$\mathcal{I}_{n,\sigma} = \mathcal{Q}_{\text{per}}(E_n \mathcal{K}_\sigma E_n) - E_n \mathcal{Q}_{\text{per}}(\mathcal{K}_\sigma) E_n.$$

RH-48 expressed this defect as an exact dyadic sum and proved that a mixed directional gain $O(\sigma^{-\gamma})$ gives

$$\|\mathcal{I}_{n,\sigma}\|_{\mathfrak{S}_2} = O(n^{-2}\sigma^{-3-\gamma}). \quad (1)$$

The strict bulk-square mesh range survives for $\gamma \leq 1/2$ [11]. RH-49 replaced the mixed gain by a purely Hilbert–Schmidt gain $\mathcal{F}_{n,\sigma}$ at a sharp quarter-power stable-rank cost. Thus $\mathcal{F} = O(\sigma^{-\delta})$ gives

$$\|\mathcal{I}_{n,\sigma}\|_{\mathfrak{S}_2} = O(n^{-2}\sigma^{-13/4-\delta}), \quad \delta \leq \frac{1}{4} \quad (2)$$

as the full-range threshold [7].

RH-50 converted the two normalized reduced-resolvent actions into left and right Hardy energies after exact Perron/parity removal [9]. RH-52 proved that the explicit range-restricted residues are $O(1)$ from weak intrinsic factors and established

$$\|B\|_{\mathfrak{S}_2}, \|C\|_{\mathfrak{S}_2} = \Theta(h\sigma^{-3/2}) \quad (3)$$

along the cell-average schedules [8]. RH-51 and RH-53 then replaced fixed-step contraction and a fitted time tail by growing-horizon, deterministic all-column certificates [10, 13]. The last uncomposed finite-matrix interface was that a sparse/full Gaussian perturbation changes the Riesz terms and the normalized coupling sources themselves.

This paper gives that composition. Its role is deliberately narrower than an unconditional Stage-A4 theorem. The exact algebra determines which Riesz-conditioning and Hardy-budget premises suffice, how their exponents combine, and how a finite certificate transfers. It does not derive the remaining uniform premises from five computed noise levels.

Main results

1. Normalizing a nonzero Hilbert–Schmidt coupling costs at most twice its relative perturbation. In combination with (3), the adaptive Gaussian cutoff is harmless on the normalized coupling ranges.
2. Common isolating contours give separate perturbation ledgers for the unweighted projectors and weighted Riesz terms. These feed directly into the two intrinsic Hardy triples.
3. The exact power telescoping identity, evaluated with actual finite-time norm ledgers, transfers a contracting block and gives a complete perturbed Hardy-energy upper.

4. The RH-48–RH-53 chain composes with the exact exponent $\delta = \alpha_B + \alpha_C$. The quarter-power threshold is therefore a budget shared by the two Hardy directions, not a quarter power available independently to each.
5. A two-dimensional nonnormal family has matrix perturbation tending to zero while its peripheral Riesz projector changes by order one. Thus cutoff convergence alone cannot close the intrinsic-factor interface.

2 Intrinsic directional triples

Let T be a fine matrix and let U and W be the coarse and Haar-detail isometries. Relative to the adjacent split,

$$T = \begin{pmatrix} T_c & B \\ C & D \end{pmatrix}, \quad B = U^*TW, \quad C = W^*TU.$$

For the two peripheral branches, write

$$\begin{aligned} P_f &= P_{f,+} + P_{f,-}, & W_f &= \lambda_{f,+}P_{f,+} + \lambda_{f,-}P_{f,-}, \\ P_c &= P_{c,+} + P_{c,-}, & W_c &= \lambda_{c,+}P_{c,+} + \lambda_{c,-}P_{c,-}. \end{aligned}$$

Here P denotes an unweighted Riesz projection and W its weighted term. Put

$$Q_f = I - P_f, \quad N_f = T - W_f, \quad Q_c = I - P_c, \quad N_c = T_c - W_c.$$

For a fixed Hardy radius r above the two bulk spectral radii, the RH-53 triples are

$$A_B = r^{-1}N_f, \quad X_B = Q_f U \hat{B}, \quad Y_B = U^*, \quad (4)$$

$$A_C = r^{-1}N_c^*, \quad X_C = \hat{C}^*, \quad Y_C = Q_c^*, \quad (5)$$

where

$$\hat{B} = B / \|B\|_{\mathfrak{S}_2}, \quad \hat{C} = C / \|C\|_{\mathfrak{S}_2}.$$

Their directional energies are

$$\mathcal{E}(A, X, Y)^2 = \sum_{m \geq 0} \|Y A^m X\|_{\mathfrak{S}_2}^2. \quad (6)$$

We compare these quantities with a second matrix, marked by tildes. All norms below are Euclidean operator norms except where $\mathfrak{S}_2$ is shown.

3 Normalization and intrinsic factor transfer

Lemma 3.1 (Normalized Hilbert–Schmidt coupling). *Let $B, \tilde{B} \neq 0$ and suppose*

$$\|B - \tilde{B}\|_{\mathfrak{S}_2} \leq \varepsilon_B < \|B\|_{\mathfrak{S}_2}.$$

Then

$$\boxed{\left\| \hat{B} - \hat{\tilde{B}} \right\|_{\mathfrak{S}_2} \leq \frac{2\varepsilon_B}{\|B\|_{\mathfrak{S}_2}}.} \quad (7)$$

The same statement holds for C .

Proof. Insert $\tilde{B}/\|B\|_{\mathfrak{S}_2}$ and use

$$\left| \left\| \tilde{B} \right\|_{\mathfrak{S}_2} - \|B\|_{\mathfrak{S}_2} \right| \leq \left\| \tilde{B} - B \right\|_{\mathfrak{S}_2}.$$

The two resulting terms are each at most $\varepsilon_B/\|B\|_{\mathfrak{S}_2}$. The strict premise also guarantees $\tilde{B} \neq 0$. $\square$

Corollary 3.2 (Adaptive coupling normalization). *Suppose (3) holds and the adaptive full-versus-sparse matrix defect is*

$$\varepsilon_h = O\left(\frac{h^2}{(\log(1/h))^{1/4}}\right).$$

The induced Haar coupling defects are at most ε_h , and hence

$$\left\| \hat{B} - \hat{\tilde{B}} \right\|_{\mathfrak{S}_2} + \left\| \hat{C} - \hat{\tilde{C}} \right\|_{\mathfrak{S}_2} = O\left(\frac{h\sigma^{3/2}}{(\log(1/h))^{1/4}}\right). \quad (8)$$

Proof. Orthogonal block extraction is contractive in Hilbert–Schmidt norm. Apply Lemma 3.1 and divide the adaptive cutoff rate of RH-39 by (3) [6]. $\square$

The remaining factors require spectral conditioning. Let Γ_+ and Γ_- be common positively oriented isolating contours for T and $\tilde{T}$. Set

$$M_s = \sup_{z \in \Gamma_s} \left\| (z - T)^{-1} \right\|, \quad \widetilde{M}_s = \sup_{z \in \Gamma_s} \left\| (z - \tilde{T})^{-1} \right\|.$$

Proposition 3.3 (Conditioned Riesz ledger). *If $\delta_T = \|T - \tilde{T}\|$, then the rank-two unweighted and weighted defects satisfy*

$$\varepsilon_P := \left\| P - \tilde{P} \right\| \leq \frac{\delta_T}{2\pi} \sum_{s \in \{+, -\}} |\Gamma_s| M_s \widetilde{M}_s, \quad (9)$$

$$\varepsilon_W := \left\| W - \widetilde{W} \right\| \leq \frac{\delta_T}{2\pi} \sum_{s \in \{+, -\}} |\Gamma_s| \max_{z \in \Gamma_s} |z| M_s \widetilde{M}_s. \quad (10)$$

Proof. Use the Riesz formulas

$$P_s = \frac{1}{2\pi i} \int_{\Gamma_s} (z - T)^{-1} dz, \quad W_s = \frac{1}{2\pi i} \int_{\Gamma_s} z(z - T)^{-1} dz,$$

and the exact resolvent identity

$$(z - T)^{-1} - (z - \tilde{T})^{-1} = (z - T)^{-1}(T - \tilde{T})(z - \tilde{T})^{-1}.$$

Sum the two contour estimates. This is the standard nonnormal conditioning dependence, retained rather than suppressed [1, 3]. $\square$

Theorem 3.4 (Factor-aware intrinsic triple transfer). *For the left triple (4), suppose*

$$\begin{aligned} \left\| T - \tilde{T} \right\| &\leq \delta_{T,f}, & \left\| W_f - \widetilde{W}_f \right\| &\leq \varepsilon_{W,f}, \\ \left\| P_f - \tilde{P}_f \right\| &\leq \varepsilon_{P,f}, & \left\| B - \tilde{B} \right\|_{\mathfrak{S}_2} &\leq \delta_B < \|B\|_{\mathfrak{S}_2}. \end{aligned}$$

Then

$$\delta_{A,B} := \|A_B - \tilde{A}_B\| \leq r^{-1}(\delta_{T,f} + \varepsilon_{W,f}), \quad (11)$$

$$\delta_{X,B} := \|X_B - \tilde{X}_B\|_{\mathfrak{S}_2} \leq \varepsilon_{P,f} + \|\tilde{Q}_f\| \frac{2\delta_B}{\|B\|_{\mathfrak{S}_2}}, \quad (12)$$

$$\delta_{Y,B} = 0. \quad (13)$$

For the right triple (5),

$$\delta_{A,C} \leq r^{-1}(\delta_{T,c} + \varepsilon_{W,c}), \quad (14)$$

$$\delta_{X,C} \leq \frac{2\delta_C}{\|C\|_{\mathfrak{S}_2}}, \quad (15)$$

$$\delta_{Y,C} \leq \varepsilon_{P,c}. \quad (16)$$

Proof. For the bulk,

$$N_f - \tilde{N}_f = (T - \tilde{T}) - (W_f - \tilde{W}_f),$$

which gives (11); adjunction gives (14). For the left source, insert $\tilde{Q}_f U \hat{B}$:

$$\left\| Q_f U \hat{B} - \tilde{Q}_f U \hat{B} \right\|_{\mathfrak{S}_2} \leq \|Q_f - \tilde{Q}_f\| + \|\tilde{Q}_f\| \left\| \hat{B} - \tilde{B} \right\|_{\mathfrak{S}_2}.$$

Use Lemma 3.1. The left observation U^* is fixed. The right source is only the normalized coupling, while its observation is the adjoint complement; this proves (15)–(16). $\square$

Remark 3.5 (Where strong–weak perturbation may enter). Equations (9)–(10) are one fully explicit route, but they may be too expensive in L^2 because the peripheral residues themselves grow logarithmically [12]. A Keller–Liverani strong–weak modulus may replace them if it yields the same two numerical outputs ε_P and ε_W [4]. The subsequent theorem uses only those outputs; it does not assume how they were obtained.

4 Growing-horizon robustness

Let (A, X, Y) and $(\tilde{A}, \tilde{X}, \tilde{Y})$ be either pair of triples. Put

$$a_j = \|A^j\|, \quad \tilde{a}_j = \|\tilde{A}^j\|, \quad \delta_A = \|A - \tilde{A}\|.$$

Theorem 4.1 (Finite directional and block transfer). *For $m \geq 1$, define*

$$d_m = \delta_A \sum_{j=0}^{m-1} a_{m-1-j} \tilde{a}_j, \quad d_0 = 0. \quad (17)$$

Then

$$\|A^m - \tilde{A}^m\| \leq d_m. \quad (18)$$

Moreover, for $\delta_X = \|X - \tilde{X}\|_{\mathfrak{S}_2}$ and $\delta_Y = \|Y - \tilde{Y}\|$,

$$\begin{aligned} & \left\| Y A^m X - \tilde{Y} \tilde{A}^m \tilde{X} \right\|_{\mathfrak{S}_2} \leq b_m, \\ & b_m := \delta_Y a_m \|X\|_{\mathfrak{S}_2} + \|\tilde{Y}\| d_m \|X\|_{\mathfrak{S}_2} + \|\tilde{Y}\| \tilde{a}_m \delta_X. \end{aligned} \quad (19)$$

If E_M and $\tilde{E}_M$ are the finite energies through time $M - 1$, then

$$|E_M - \tilde{E}_M| \leq D_M := \left(\sum_{m=0}^{M-1} b_m^2 \right)^{1/2}. \quad (20)$$

Finally, if $q_M = \|A^M\|$ and

$$\boxed{q_M + d_M < 1}, \quad (21)$$

then $\|\tilde{A}^M\| < 1$ and the perturbed full energy obeys

$$\boxed{\mathcal{E}(\tilde{A}, \tilde{X}, \tilde{Y})^2 \leq (E_M + D_M)^2 + \|\tilde{Y}\|^2 \frac{(q_M + d_M)^2}{1 - (q_M + d_M)^2} \|\tilde{X}\|_{\mathfrak{S}_2}^2 \sum_{j=0}^{M-1} \tilde{a}_j^2}. \quad (22)$$

Proof. The exact identity

$$A^m - \tilde{A}^m = \sum_{j=0}^{m-1} A^{m-1-j} (A - \tilde{A}) \tilde{A}^j$$

gives (18). Insert and subtract $\tilde{Y} A^m X$ and $\tilde{Y} \tilde{A}^m X$ to obtain (19); Minkowski in the finite time ℓ^2 space proves (20). The power bound at $m = M$ gives (21). The block-geometric tail from RH-53 gives

$$\sum_{m \geq M} \|\tilde{Y} \tilde{A}^m \tilde{X}\|_{\mathfrak{S}_2}^2 \leq \|\tilde{Y}\|^2 \frac{\|\tilde{A}^M\|^2}{1 - \|\tilde{A}^M\|^2} \sum_{j=0}^{M-1} \|\tilde{A}^j \tilde{X}\|_{\mathfrak{S}_2}^2.$$

Use $\|\tilde{A}^M\| \leq q_M + d_M$ and $\|\tilde{A}^j \tilde{X}\|_{\mathfrak{S}_2} \leq l e \tilde{a}_j \|\tilde{X}\|_{\mathfrak{S}_2}$. □

Remark 4.2 (Why the finite ledger matters). Replacing a_j and $\tilde{a}_j$ by $\|A\|^j$ and $\|\tilde{A}\|^j$ can make (21) useless for a nonnormal bulk even when an actual block power is strongly contracting. The theorem stores the measured or enclosed transient through the selected growing horizon. It does not infer transient behavior from spectral radius or one-step norm.

5 A nonnormal no-free-lunch theorem

It remains to ask whether the adaptive matrix cutoff bound alone can make the Riesz defects in Theorem 3.4 vanish. The answer is no without a conditioning premise.

Proposition 5.1 (Small operator error does not control Riesz factors). *Fix $c > 0$ and let*

$$T_K = \begin{pmatrix} 0 & K \\ 0 & 1 \end{pmatrix}, \quad E_K = \begin{pmatrix} 0 & 0 \\ c/K & 0 \end{pmatrix}.$$

Then $\|E_K\| = c/K \rightarrow 0$. The eigenvalues of $\tilde{T}_K = T_K + E_K$ are

$$\lambda_{\pm} = \frac{1 \pm \sqrt{1 + 4c}}{2},$$

independent of K . Let P_K be the Riesz projector of T_K at 1 and $\tilde{P}_K$ the Riesz projector of $\tilde{T}_K$ at λ_+ . Then

$$\liminf_{K \rightarrow \infty} \|P_K - \tilde{P}_K\| \geq \frac{\sqrt{1+4c}-1}{2\sqrt{1+4c}} > 0. \quad (23)$$

Consequently there is no dimension- and conditioning-free Lipschitz bound $\|P - \tilde{P}\| \leq C \|T - \tilde{T}\|$ for nonnormal families.

Proof. The characteristic polynomial of $\tilde{T}_K$ is $\lambda^2 - \lambda - c$. Since the two eigenvalues are distinct,

$$P_K = T_K, \quad \tilde{P}_K = \frac{\tilde{T}_K - \lambda_- I}{\lambda_+ - \lambda_-}.$$

The $(1, 1)$ entry of $P_K - \tilde{P}_K$ has modulus

$$\frac{-\lambda_-}{\lambda_+ - \lambda_-} = \frac{\sqrt{1+4c}-1}{2\sqrt{1+4c}},$$

which lower-bounds the operator norm. Meanwhile $\|E_K\| = c/K$. $\square$

For $c = 1/4$, the lower bound is about 0.1464, while a numerical evaluation at $K = 10^6$ gives a projector defect about 2.93×10^5 because the off-diagonal projector geometry is itself highly conditioned. The fixed entry lower bound is enough for the theorem; the stronger growth illustrates why recomputed finite factors cannot be certified from an unweighted cutoff norm alone.

Remark 5.2. The proposition does not say that the folded-Gaussian Riesz terms are unstable. It says that their stability must be proved from their own spectral geometry—for example, [Proposition 3.3](#) or a strong-weak transfer theorem. The five-scale audit below observes excellent stability, but observation is not a substitute for this premise.

6 Conditional closure of intrinsic identification

The factor-aware finite-matrix theorem can now be joined to the earlier range-resolvent reductions. Let the residue terms be the normalized fine/coarse actions entering RH-50. RH-52 proves they are uniformly bounded along the adopted cell-average schedules.

Condition 6.1 (Dyadic factor-aware Hardy budget). There are fixed $r < \min_s \inf_{z \in \Gamma_s} |z|$, exponents $\alpha_B, \alpha_C \geq 0$, and $C < \infty$ such that, at every required dyadic level,

$$\text{spr}(N_f), \text{spr}(N_c) < r, \quad (24)$$

$$\mathcal{E}_B(r) \leq C\sigma^{-\alpha_B}, \quad \mathcal{E}_C(r) \leq C\sigma^{-\alpha_C}, \quad (25)$$

$$\text{all normalized peripheral range residues} \leq C. \quad (26)$$

The same condition may use fixed powers of $\log(1/\sigma)$ instead of power laws. Sparse certificates may be transferred to the full family by [Theorems 3.4](#) and [4.1](#), provided their Riesz ledgers and contraction margins obey those theorems uniformly along the dyadic chain.

Theorem 6.2 (RH-48–RH-53 composition). *Under [Condition 6.1](#), the purely Hilbert–Schmidt directional gain of RH-49 satisfies*

$$\sup_{j \geq 0} \mathcal{F}_{2^j n, \sigma} = O(\sigma^{-\delta}), \quad \delta = \alpha_B + \alpha_C. \quad (27)$$

Consequently

$$\boxed{\|\mathcal{I}_{n,\sigma}\|_{\mathfrak{S}_2} = O\left(n^{-2}\sigma^{-13/4-\alpha_B-\alpha_C}\right)}. \quad (28)$$

If

$$\alpha_B + \alpha_C \leq \frac{1}{4}, \quad (29)$$

then every strict schedule $n(\sigma)\sigma^2 \rightarrow \infty$ retains the intrinsic bulk-square conclusion. If both energies are polylogarithmic, then for some fixed a ,

$$\|\mathcal{I}_{n,\sigma}\|_{\mathfrak{S}_2} = O\left(n^{-2}\sigma^{-13/4}(\log(1/\sigma))^a\right). \quad (30)$$

Proof. The two-pole Laurent decomposition of RH-50 writes each normalized Hilbert–Schmidt range action as finitely many residue terms plus a bulk term. The residues are $O(1)$ by (26). The directional Hardy theorem gives

$$\begin{aligned} \ell_B^{(2)} &\leq C\{1 + \mathcal{E}_B(r)\}, \\ \ell_C^{(2)} &\leq C\{1 + \mathcal{E}_C(r)\}, \end{aligned}$$

where the fixed factor $(d_{\Gamma}^2 - r^2)^{-1/2}$ is absorbed in C . The RH-49 gain is a finite branch sum of products, so

$$\mathcal{F}_{n,\sigma} \leq C(1 + \mathcal{E}_B(r))(1 + \mathcal{E}_C(r)). \quad (31)$$

This proves (27). The quarter-power stable-rank bridge adds $1/4$ to the mixed-gain exponent, and (1) then gives (28). The RH-49 threshold is precisely $\delta \leq 1/4$. Products of fixed logarithmic powers remain a fixed logarithmic power, proving (30). $\square$

Remark 6.3 (What has and has not closed). The theorem closes the logical composition and the factor-aware finite-matrix transfer. It does not prove Condition 6.1. In particular, five finite noise levels do not give a dyadically uniform analytic Hardy trace budget, and an adaptive cutoff norm does not by itself give the Riesz modulus required by Theorem 3.4. Stage A4 is therefore a conditional closure criterion, not an unconditional small-noise identification theorem.

7 Five-scale transfer audit

The numerical experiment deliberately uses the smaller resolution

$$N\sigma = 5.12, \quad r = 0.85,$$

so that full dense Gaussian matrices and all intrinsic factors can be recomputed. The noise levels are 0.16, 0.08, 0.04, 0.02, 0.01, the dimensions are 32, 64, 128, 256, 512, and the horizons are 4, 8, 16, 24, 32. For every scale we build the full matrix and three row-renormalized sparse matrices with declared cutoffs $L = 5, 6, 8$. The $L = 5$ family is an intentionally enlarged stress test, not the production choice.

The audit records the Markov and Haar coupling defects, recomputed unweighted and weighted Riesz defects, bulk and normalized-source defects, actual semigroup-power differences, the telescoping uppers of Theorem 4.1, exact Lyapunov energies, deterministic all-column certificates, and the complete transferred upper (22). Every quantity in this section is binary64.

σ	N	M	$\ \Delta T\ _2$	$\ \Delta W_f\ _2$	$\ \Delta \hat{B}\ _{\mathfrak{S}_2}$	left margin ratio	right margin ratio
0.16	32	4	2.9290×10^{-8}	2.3658×10^{-8}	1.8709×10^{-7}	4.1067×10^{-8}	3.5197×10^{-8}
0.08	64	8	4.9913×10^{-8}	3.7163×10^{-8}	3.4759×10^{-7}	1.1302×10^{-7}	9.8658×10^{-8}
0.04	128	16	4.8876×10^{-8}	2.3020×10^{-8}	3.6353×10^{-7}	1.6693×10^{-7}	1.5328×10^{-7}
0.02	256	24	5.5817×10^{-8}	2.9979×10^{-8}	3.6935×10^{-7}	2.7219×10^{-7}	2.4017×10^{-7}
0.01	512	32	6.6073×10^{-8}	2.6251×10^{-8}	4.1159×10^{-7}	3.3308×10^{-7}	2.9933×10^{-7}

Table 1: Five-sigma stress ledger. A margin ratio is $d_M/(1 - q_M)$. All transferred blocks remain contractive with overwhelming margin.

The largest Markov spectral defect is 6.61×10^{-8} , the largest fine weighted-Riesz defect is 3.72×10^{-8} , and the largest normalized B defect is 4.12×10^{-7} . Every composed factor upper dominates the directly recomputed triple defect. The largest transferred block perturbation consumes only 3.34×10^{-7} of the available contraction margin; all actual block norms stay between 0.0067 and 0.031.

σ	left actual ΔE^2	left finite upper	left full E	left transferred	right actual ΔE^2	right finite upper	right transferred
0.16	4.47×10^{-9}	1.28×10^{-6}	0.90396	0.90422	2.11×10^{-12}	1.00×10^{-6}	1.00299
0.08	9.65×10^{-9}	7.47×10^{-6}	1.16256	1.16387	3.50×10^{-8}	7.88×10^{-6}	1.26674
0.04	1.36×10^{-7}	3.36×10^{-5}	1.33383	1.33491	1.86×10^{-7}	3.34×10^{-5}	1.48583
0.02	1.48×10^{-7}	1.35×10^{-4}	1.40958	1.41156	3.42×10^{-7}	1.46×10^{-4}	1.63590
0.01	1.70×10^{-7}	5.14×10^{-4}	1.46807	1.46932	4.93×10^{-7}	5.81×10^{-4}	1.76159

Table 2: Hardy-energy transfer at $L = 5$. “Actual” compares dense Lyapunov energies; “finite upper” is the factor-aware main-sum perturbation; “transferred” uses (22) without a full Lyapunov solve.

At $L = 6$, the Markov defect is about 10^{-10} from dimension 64 onward; at $L = 8$, direct differences are at binary64 roundoff. On these pilot dimensions the sufficient adaptive prescription is still $L(h) = 5$. Its distinction from a fixed window is asymptotic: RH-39 proves that the adaptive multiple eventually grows like $2\sqrt{\log(1/h)}$, whereas any fixed multiple has a positive row-norm continuum floor.

Factor-aware sparse-to-full transfer for intrinsic Hardy triples

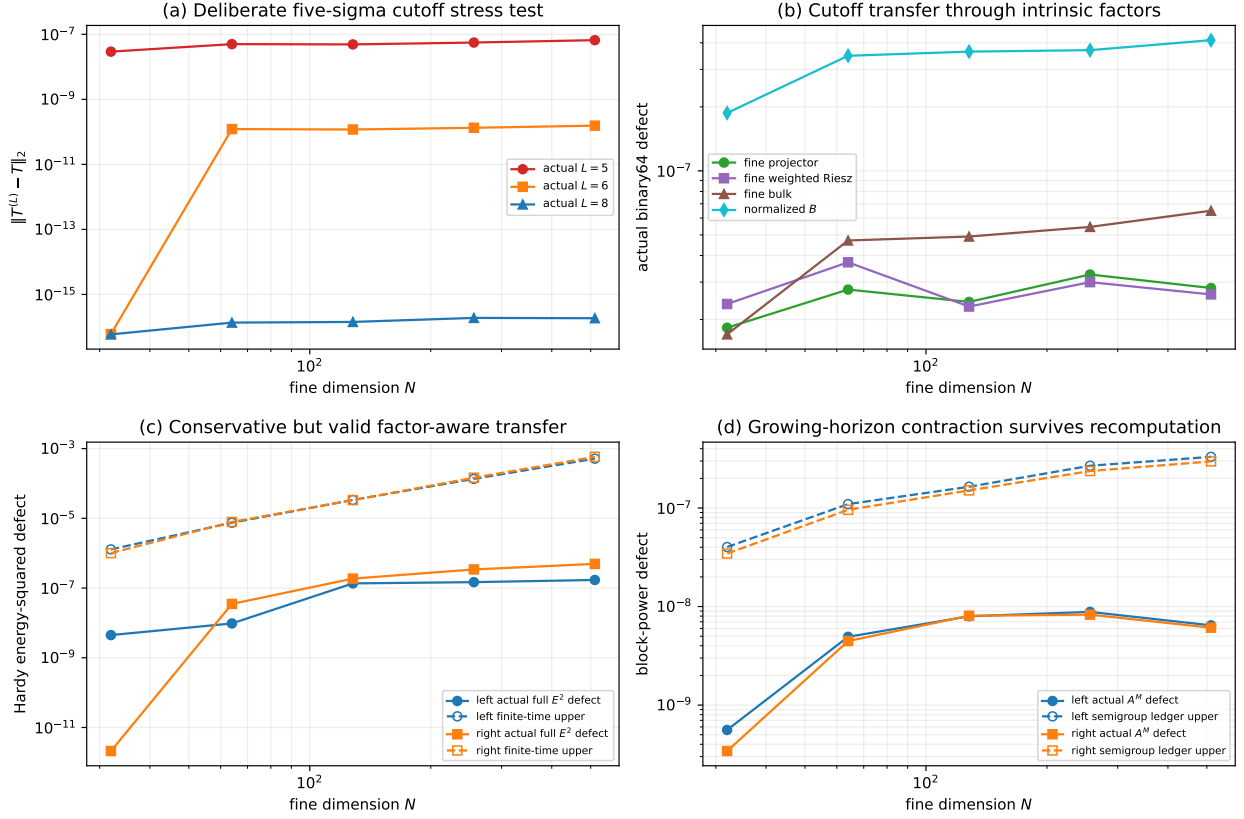


Figure 1: Factor-aware sparse-to-full audit. (a) The declared five-sigma window creates a visible stress defect; six and eight sigma rapidly reach roundoff. (b) Recomputed intrinsic factors and the normalized outgoing coupling. (c) Actual Hardy-energy changes and conservative finite-time uppers. (d) Actual growing-block defects and the semigroup telescoping ledgers. All panels are binary64 diagnostics.

7.1 Outward-rounded arithmetic audit

A separate 256-bit Arb execution encloses a 3×2 coupling, its perturbation, both normalizations, a factor-aware triple ledger, and six semigroup powers [2, 5]. It certifies

$$\left\| \Delta \hat{B} \right\|_{\mathfrak{G}_2} \leq 1.251468 \times 10^{-8}, \quad q_6 + d_6 \leq 0.025000445.$$

This validates outward arithmetic for the formulas. It is intentionally a small abstract execution and not an interval eigensolver or production-scale Riesz enclosure.

8 Theorem boundary and next gate

The status after this composition is as follows.

Normalized coupling gate

Closed analytically. The adaptive cutoff becomes even smaller after division by the physical coupling scale.

Factor-aware finite transfer

Closed analytically once projector and weighted-Riesz defect uppers are supplied. A common-contour ledger is explicit.

Growing-horizon robustness

Closed analytically. The certificate uses actual finite semigroup ledgers and transports both contraction and the full Hardy upper.

RH-48–RH-53 synthesis

Closed as a conditional theorem with exact exponent budget $\alpha_B + \alpha_C \leq 1/4$.

Uniform Stage A1 budget

Open. No analytic theorem yet bounds both dyadic Hardy energies uniformly, polylogarithmically, or with a proved total power at most $1/4$.

Uniform intrinsic Riesz modulus

Open. The dense audit observes stability, but the no-go theorem shows that an operator cutoff bound alone cannot prove it.

Unconditional Stage A4

Open. The preceding two premises prevent an unconditional claim despite the complete finite-dimensional transfer architecture.

The sharp next target is no longer another fixed-window pilot. One needs a dyadically uniform analytic Hardy/Stein trace budget together with either a strong–weak intrinsic-factor modulus or an outward-rounded contour ledger on the production family. If these provide total Hardy exponent at most $1/4$, [Theorem 6.2](#) closes the identification theorem immediately. If every valid budget exceeds $1/4$, the full strict $n\sigma^2 \rightarrow \infty$ range fails and a stronger mesh schedule must be stated instead.

No result here constructs an arithmetic trace formula, a prime-power identity, a self-adjoint Hilbert–Pólya operator, a $T \log T$ counting law, or a zeta-zero spectral identity, and no Riemann-hypothesis conclusion is drawn. The independent TPC twin-prime branch is not a premise of this theorem.

9 Reproducibility

The archive contains theorem code in `src/intrinsic_transfer`, the five-scale pilot, the Arb audit, tests, figure sources, a machine-readable closure certificate, dependency hashes, and the manuscript. From this directory, the main replay is

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/pytest -q
OPENBLAS_NUM_THREADS=16 OMP_NUM_THREADS=16 \
  PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/run_factor_aware_pilot.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/run_arb_transfer_audit.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_closure_certificate.py
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
```

```
experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

10 Conclusion

The sparse/full gap left by deterministic Hardy tails can be propagated through recomputed intrinsic factors without choosing eigenvector gauges. Normalized Haar couplings are stable at twice their relative Hilbert–Schmidt error, the weighted and unweighted Riesz terms enter through separate auditable defects, and one growing contracting block survives under the exact finite semigroup ledger. These ingredients compose RH-48 through RH-53 into a single closure theorem with the transparent threshold $\alpha_B + \alpha_C \leq 1/4$.

The result also marks the remaining wall precisely. Nonnormal Riesz stability is not free, and finite Hardy plateaus do not prove a uniform trace budget. Thus the route remains viable and substantially clearer, but its last analytic premises must still be proved or validated before the small-noise intrinsic identification can be called unconditional.

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